Contact

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www.linkedin.com/in/wasim-aftab (LinkedIn)
github.com/wasimaftab (Other)
wasimaftab.de/ (Portfolio)

Top Skills

Bioinformatics

Research Software Engineering

Multi-Omics Data Analysis

Languages

English (Full Professional)

Hindi (Full Professional)

Bengali (Full Professional)

Arabic (Limited Working)

German (Limited Working)

Certifications

Quantitative proteomics: Strategies and tools to probe biology

Python for Genomic Data Science

Statistics for Genomic Data Science

Introduction to Genomic Technologies

EMBO | EMBL Symposium: Multiomics to Mechanisms: Challenges in Data Integration

Publications

Molecular Wiring of a Mitochondrial Translational Feedback Loop

MALDI-IMS combined with shotgun proteomics identify and localize new factors in male infertility

Molecular Connectivity of Mitochondrial Gene Expression and OXPHOS Biogenesis

On Classification of PDZ Domains: A Computational Study

A Weighted Cosine RBF Neural Networks

Wasim Aftab

Computational Biologist & Research Software Engineer | Multi-Omics, Single-Cell & Biomedical AI | Python, R, Linux/HPC

Neuried, Bavaria, Germany

Summary

I am a computational biologist and research software engineer with a PhD in Bioinformatics and more than 10 years of experience turning complex biomedical data into reproducible analyses and usable software.My work spans transcriptomics, epigenomics, proteomics, spatial omics and medical imaging. At LMU, I have developed workflows for bulk and single-cell RNA-seq and ATAC-seq, statistical methods for proteomics, and machine-learning pipelines for multimodal MRI. I enjoy working across the full process: shaping a biological question with collaborators, designing the analysis, building versioned workflows for HPC, and translating the results into clear biological conclusions.Before moving into bioinformatics, I worked in scientific programming and software engineering, developing C/Fortran code for HPC and Java-based automation systems. That background still shapes how I work: methods should be statistically sound, software should be maintainable, and analyses should be reproducible.Current interests include multi-omics integration, single-cell and spatial analysis, biomedical machine learning, research software, and knowledge-grounded AI for science.Python | R | Snakemake | Linux/HPC | Docker | PyTorch | Bioconductor | FastAPI | Neo4j

Experience

LMU Klinikum München

Postdoctoral Researcher

May 2025 - Present (1 year 4 months)

Munich

• Develop computational methods that integrate single-cell reference atlases, bulk proteomics and medical imaging to study neurovascular disease.

• Created HASA, a hierarchy-aware method for attributing bulk proteomic signals to cell types using single-cell RNA-seq reference data and replicate-aware stability estimates.

• Validated HASA using a mouse cerebral amyloid angiopathy dataset, identifying smooth-muscle-cell-specific changes and corresponding biological pathways, and packaged the method as an installable application for collaborators.
• Built SiderUNet, a reproducible 3D nnU-Net pipeline for segmenting cortical superficial siderosis from multimodal MRI, achieving an out-of-fold Dice score of 0.75.
• Designed an uncertainty-driven active-learning workflow that improved annotation quality from a mean Dice score of 0.56 to 0.81 and reduced approximately three hours of manual annotation per case.
• Train and deploy models on GPU and HPC systems and maintain reproducible Python pipelines.
• Contribute to weekly MSc project meetings by helping shape analytical tasks and providing methodological feedback and suggestions on students' results alongside the project supervisor.

Biomedical Center (BMC) LMU Munich

9 years 8 months

Postdoctoral Researcher

January 2022 - April 2025 (3 years 4 months)
Munich

• Built and maintained reproducible Snakemake workflows for bulk RNA-seq, single-cell RNA-seq and ATAC-seq, reducing data-processing time by approximately 50% and standardising results across research projects.
• Worked with chromatin-biology groups within SFB 1064 from study design and quality control through statistical analysis and biological interpretation.
• Developed statistical R pipelines for high-throughput proteomics, including batch-effect correction, missing-value handling, exploratory analysis and moderated inference.
• Created a biomedical question-answering application using Python, FastAPI and JavaScript. Its keyword-frequency prompt-enhancement method improved information retrieval by 137.5% in the study evaluation and was published in BMC Bioinformatics.
• Prototyped knowledge-grounded AI workflows using GPT-4, Neo4j and GraphRAG for biomedical entity extraction, relationship discovery and traceable evidence retrieval.
• Deployed research applications using Docker and Nginx and collaborated closely with experimental researchers to translate computational results into biological conclusions.

Research Assistant during PhD
September 2015 - December 2021 (6 years 4 months)
Germany

• Developed ImShot, open-source software that integrates MALDI imaging mass spectrometry with shotgun proteomics for probabilistic in-situ protein identification and visualisation.
• Designed statistical methods and reproducible R pipelines for proteomics, protein-complex discovery and gene-expression network analysis.
• Built R packages, Shiny applications and Electron-based research software used by more than ten collaborating laboratories.
• Contributed computational analyses and software to interdisciplinary studies of mitochondrial gene expression, OXPHOS biogenesis and spatial proteomics.
• Worked directly with experimental researchers to translate biological questions into statistical analyses, usable software and peer-reviewed publications.

iCelero Technologies Pvt. Ltd.
Senior Software Engineer
April 2012 - December 2012 (9 months)
Bengaluru

• Developed automated software-testing workflows using Java, Selenium and AutoIt, substantially reducing repetitive manual testing.
• Translated application requirements and test scenarios into reusable automation components and maintainable quality-assurance workflows.

Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR)
Scientific Programmer
August 2009 - March 2012 (2 years 8 months)
Greater Bengaluru Area

• Optimised C and Fortran linear-algebra routines for quantum-chemistry simulations running on high-performance computing systems.
• Administered and troubleshot Linux workstations used by computational research groups.
• Developed a C program for predicting the functions of hypothetical proteins from Protein Data Bank structural features.

Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR)
Research Internship

March 2009 - June 2009 (4 months)
Bengaluru

During the internship, I worked on my final year Bachelor's degree thesis, which involved developing a C program to diagonalize large sparse matrices, a crucial tool for solving quantum chemistry problems, particularly those involving large-scale configuration interaction calculations of electronic wave functions.

Education

Ludwig-Maximilians-Universität München

Doctor of Philosophy - PhD, Bioinformatics ( ) · (September 2015 - December 2021)

King AbdulAziz University

MSc, Computer Engineering ( ) · (2013 - 2015)

Visvesvaraya Technological University

Bachelor of Engineering, Computer Engineering , ( ) · (2005 - 2009)